AI-Powered Payment Gateway Platform

Case study · DL4 · 10-minute presentation (≈ 7:25 spoken + Q&A)

Labels used on every slide

- **Done**: code and a passing test (Status "Done" in `docs/requirements-traceability.md`)
- **Partial**: only part is built; the missing part is named on the slide
- **Design only**: described in `docs/design.md`; no code and no measurement

Every number comes from `README.md`, `docs/sample-queries.md`, `docs/requirements-traceability.md` or `docs/perf/README.md`.

1. Problem and scope

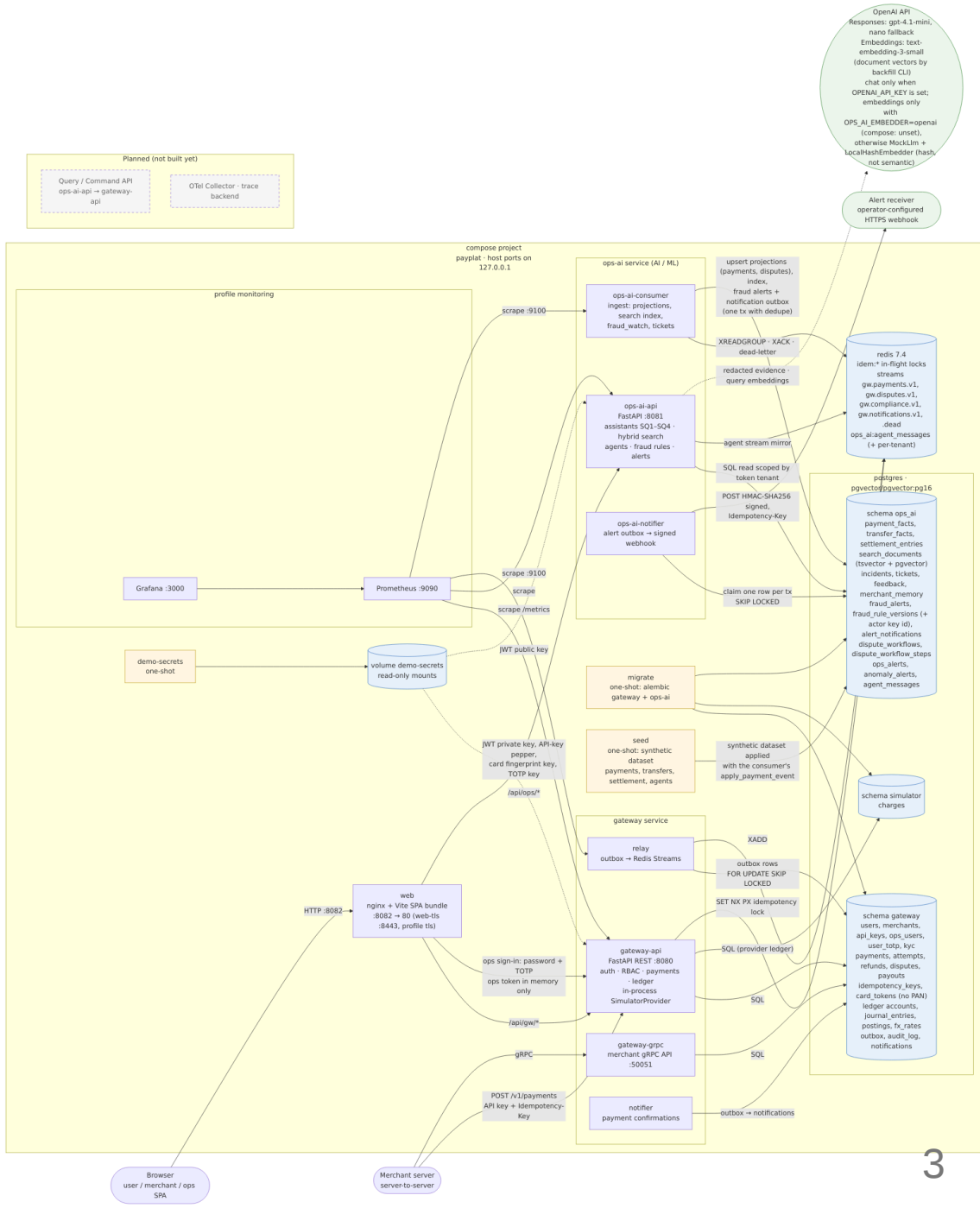
Goal: when a payment fails, is declined, looks fraudulent, or a payout is lower than expected, merchants and ops staff get the **cause, the evidence and a fix**, while the payment path stays correct.

- Correctness bar: no double charge, the ledger balances, card numbers are never stored, it keeps working with the LLM down
- Scope: the brief's **Basic 13** items + **SQ1–SQ4** + **NFR**
- Not done: real money movement, real card network, real KYC; a **simulator and seed data** stand in
- Data: synthetic, **seed 42**, **12,000 records + 5 incidents**

2. Architecture and key decisions

- **Two services**: `gateway` (payments, ledger, auth) and `ops-ai` (projection, search, assistant)
- They share **only** `contracts/` (`PaymentEventV1`); import boundary enforced by `tests/test_import_rules.py`
- Events: transactional **outbox → relay → Redis Streams** `gw.payments.v1` → consumer; duplicates dropped via `processed_events`, malformed → dead-letter
- **Key decision**: AI stays **out of the payment write path**; if the LLM is down, charges still go through
- Known gap: Redis has **no ACL** (slide 9)

Diagram: `docs/architecture.png` (full size: `docs/architecture.pdf`; zoomed: Appendix D)



3. Core payments and `make demo`

Feature	Status	Key point
Wallet / P2P	**Done**	Double-entry ledger, minor units; `FOR UPDATE` taken in account-id order
Merchant card charge	**Done**	Tokenise first, then charge; the provider is called outside the DB transaction
Multi-currency / FX	**Done**	FX legs via per-currency clearing accounts; `settlement_currency` kept on merchant REST, consumer REST (`/v1/me/payments`) and gRPC `CreatePayment` , with round-trip tests. Static seed rates, no live feed; chargebacks not yet FX-aware
Idempotency	**Done**	A Postgres unique row decides; the Redis lock only reduces contention
REST + gRPC	**Done**	Same domain service, same validation module; 422 → gRPC status per code (reused key / zero after FX → `INVALID_ARGUMENT` , no rate → `FAILED_PRECONDITION` , other → `INTERNAL`); demo gRPC is plaintext on loopback

- Idempotency answers: retry → stored body · same key, other body → 422 `idempotency_key_reused` · in flight → 409 after 5 s · outcome unknown → **202, never "failed"**
- **Known failure**: the 20-way same-key test (P0-1) failed once under load (1 of 20 got 409); cause not identified; passes 3/3 when run alone
- `make demo` : no `.env` , no key. Seed → 3 live charges (approved / declined / same-key retry) → SQ1–SQ4 with checks; **exit 0 only if every check passes** (Appendix A)

4. AI features and what is proven

- **Tools-first**: tools produce every number, label and fix; the LLM only turns evidence into text and gets no tools while narrating
- NL failure query (B4) and RAG (B5); hybrid search (B6) fuses FTS + pgvector with RRF (k=60). The default embedder `LocalHashEmbedder` is hash-based, spelling-level: **not semantic search** (needs `OPS_AI_EMBEDDER=openai` + a key; unset in compose)
- Fraud (B7): card-testing detection; on synthetic data **40/40 bursts detected, FP 2.5 %** (4 of 160 legitimate groups)
- Tool calling (B9): `/v1/assistant/query` is **one planning call (≤ 4 tools)**, not an agent loop
- Multi-agent (R2-1) and LLM-as-judge (R2-2): **Partial** · `make eval` deterministic gate runs in CI; LLM-judged metrics **skipped** without a key (P0-15 **Partial**)
- **What is proven**: the keyless demo uses **MockLlm**. It shows the tool execution and response path, **not natural-language quality**

5. SQ1: 15 % afternoon card failures, cause and fix

`POST /v1/assistant/investigate` · merchant `mch_0001` · traceability: SQ1 **Done**

- Observed: window **36 of 241 failed = 14.9 %**; 7-day baseline **7 of 390 = 1.8 %** (8.3×)
- **32 of the 36** failures are `issuer_unavailable` on BIN `400000`
- Answer: `issuer_outage_on_bin`. Fix: **reroute BIN 400000 to acquirer** `acq_b` until the issuer recovers (cites past incident `inc_0001`)
- The demo's live declined charge is part of the evidence
- Limit: the `/failures` route searches incidents by FTS and finds none on the seed; **only** `/investigate` **cites the incident**

6. SQ2: cross-border transfer declined despite the balance

`POST /v1/assistant/investigate-transfer` · seeded sender, user token, role `consumer` · traceability: SQ2

Done

- Observed: transfer **1,800.00 USD** (US → IN), balance **5,250.00 USD**: **not insufficient funds**
- Rule that fired: **KYC tier 1 cross-border limit, 1,000.00 USD per transfer** (`cross_border_limit_for_kyc_tier`, owner `sender`)
- Fix: upgrade to KYC tier 2 (limit 10,000.00 USD per transfer)
- Role checks: merchant token → **403** on this route; user token → **403** on `/investigate`
- Limit: the transfer is a **seeded decision record**, not one the gateway executed

7. SQ3: micro-payment series, card testing?

`POST /v1/assistant/investigate-fraud` · merchant `mch_0002` · traceability: SQ3 **Done**

Episode	Micro payments	Cards	Decline ratio	Result
Demo live burst	12	4	0.5	flagged (`micro_burst`, `many_cards`, `high_decline_ratio`)
Seeded burst	40	unknown	1.0	flagged (`micro_burst`, `card_identity_unknown`, `high_decline_ratio`)

- Answer: `card_testing`. Each episode raises **1 alert + 1 ticket** in the `fraud` queue (priority `high`)
- Each alert queues **1 outbound notification** in the watcher's transaction (outbox); `ops-ai-notifier` (started by `make demo`) delivers it
- Fixes: velocity limit for payments ≤ 1.00 · require 3-D Secure · do not touch the cards in the burst
- Limit: the demo wires **no notification receiver** (log sink: only a line in the `ops-ai-notifier` log); a signed webhook only when `OPS_AI_ALERT_WEBHOOK_URL` is set

8. SQ4: why the payout is lower than expected

`POST /v1/assistant/explain-settlement` · merchant `mch_0003` · 2026-10 · USD · traceability: SQ4 **Done**

Line	Amount (USD)	Payments
gross	11,251.37	202
MDR (2.90 %)	−326.29	202
fixed fees	−60.60	202
FX	−8.76	10
refunds	−283.93	8
chargebacks	−125.40	3
chargeback fees	−45.00	3
reserve (5 %)	−562.57	202
net	**9,838.82**	

- Answer: the **1,086.26 USD** gap to the expected 10,925.08 USD, line by line; checked against **202 projected payments before any text is written** (`verified: true`)
- Limit: settlement lines are a **seeded acquirer report**; the gateway's own settlement batch and payout are **Design only**

9. Security: done and not done, side by side

Done	Not yet
Card tokenisation: only token, BIN, last4 and an HMAC fingerprint are stored; the PAN is never stored, logged or published	**No Redis ACL**: anyone who reaches Redis can push fake payment events
Passwords: argon2id; at most 4 concurrent verifies, 503 after a 2 s wait	Postgres: **superuser login** + `SET ROLE` (can `RESET ROLE`)
TOTP secret sealed with AES-256-GCM, user id as AAD	No per-IP login rate limit; signup 409 reveals whether an email exists
API keys stored as HMAC-SHA256 hashes; tokens RS256 with iss / aud / exp checks	No merchant-token revocation, no key rotation (`kid`)
Secrets as `SecretStr`; each service mounts only its own secrets, read-only	**TLS only browser → nginx** (`make tls-up`, dev cert); nginx → web, web → API, Postgres, Redis plaintext on the compose network; service mTLS **Design only**

- NFR-SEC transport: **Partial** · append-only audit log (trigger + grants) built, but P0-10 **Partial** because of the superuser login

10. NFR: what was measured and what was not

k6 on `POST /v1/payments` (card). Target: **p99 < 2 s and error < 1 %** (`docs/perf/README.md`)

Offered load	Measured throughput / completed	p50	p95	p99	Errors	Target
2 req/s	181 requests, none dropped	405 ms	1.91 s	**2.82 s**	0 %	**Not met**
40 req/s	saturates at ≈ 7 req/s; 745 completed, 2,856 iterations dropped	24.9 s	41.4 s	44.5 s	0 %	**Not met**

- Environment: a **shared laptop VM** next to other compose stacks; the load hits **3 test cards**, and each card takes a lock
- **Design only** (no code, no measurement): 100k TPS / 100M users (P2-7), 99.999 % (P2-2), multi-region (P2-1)

11. Production design and crash recovery

- **99.999 %** (P2-2) covers the **payment write path** only; error budget 25.9 s/month ⇒ failover in seconds · **Design only**
- **DB choice** (P2-1): Spanner multi-region vs per-region Cloud SQL + reconciliation; money tables → Spanner · **Design only**
- **100k TPS** (P2-7): an *estimate* of 1 charge ≈ 10 row writes ⇒ ≈ 1M row writes/s; one Postgres is not enough → partition by account / merchant id · **Design only**
- **Crash matrix**: C2–C4 checked **by in-process fault injection**. C1 (gateway dies after provider success) and C5 (outcome unknown past retention) are **Design only**; no real process-kill test
- **Reconciler (design proposal)**: find provider successes missing a local record and converge them. Assumes a stable match key, a provider-side lookup and an idempotent re-run. **Not built**: not a recovery guarantee

12. Known gaps, next steps and summary

Main Partial rows (full list: Appendix B)

- Payment correctness: P0-1 (no Redis flush / lock-expiry run), P0-4 (no reconciler, no real kill)
- AI: R2-1 multi-agent, R2-2 LLM-as-judge, P0-15 (LLM-judged eval not run), P1-8 (no prompt hash, no token budget)
- Ops: P0-13 metrics, P0-14 alerts, P1-2 retry + backoff, P1-5 load test; PR-2 (alert → runbook demo) not recorded

Next steps, in this order: ① build the reconciler (converge C1/C5, resolve `unknown`) ② re-measure on a dedicated environment with many cards ③ Redis ACL + per-service DB roles ④ add a key and run LLM-judged eval in CI

Summary: SQ1–SQ4 run end to end with tool evidence and checks. The correctness core (idempotency, ledger) is pinned by tests, **with one known concurrent-test failure**. The performance target is **missed** and production scale is **unmeasured**, and both are labelled that way.

Appendix A: the 9 steps of `make demo` (`scripts/demo.py`)

1. Get an API key for seeded SQ1 merchant `mch_0001`; sign up `demo+<random>@example.com`
2. Over HTTP: 1 approved charge, 1 decline on the SQ1 BIN (`4000000000000119`, issuer unavailable), 1 retry with the approved charge's `Idempotency-Key` (identical body checked)
3. Exchange the API key for a merchant token (RS256 JWT, issuer `gateway`, audience `ops-ai`, 15 min)
4. Wait until both payments are on the stream and the consumer group has lag 0, pending 0
5. Ask "declined payments"; check the evidence cites only the declined side
6. SQ1 (`/v1/assistant/investigate`): label `issuer_outage_on_bin`, top BIN `400000`, window 10–20 % vs baseline < 5 %, `inc_0001`, live decline cited
7. SQ2 (`/v1/assistant/investigate-transfer`): set a fresh password on the seeded sender, log in; check `cross_border_limit_for_kyc_tier` and 403
8. SQ3 (`/v1/assistant/investigate-fraud`): live burst on `mch_0002`, 12 × 0.50 USD (4 cards, 6 declined); check `card_testing`, alert, `fraud` ticket cited
9. SQ4 (`/v1/assistant/explain-settlement`): `mch_0003`; check the order and sum of the 8 lines, net < expected, 403

Only the first 8 characters of the API key are printed; passwords and tokens are never printed.

Appendix B: Partial rows in the traceability matrix (1/2)

Area	Row	What is missing (from the traceability Status)
Payment correctness	P0-1	No run with Redis flushed / lock expired
	P0-2	Ledger property test is broad, but the row is Partial
	P0-4	No reconciler, no real process kill, no `tests/e2e/crash/`
	P1-1	Crash coverage depends on P0-4
	P1-2	No retry + backoff + jitter, no N-failures rule
AI	P0-5	T2 proven; no test of charges, risk flags and card-testing alerts with `LlmPort` failing
	P0-6	Card-number masking done; no all-field canary for email / phone / password / token
	P0-7	No `suggestion_id` flow, no refund with the human's JWT
	P0-8	No injection fixture in resolution notes, no test rejecting schema-invalid output
	P0-15	LLM-judged metrics skipped
	P1-8	No dated model-id pin in env, no prompt hash, no token budget
	R2-1 / R2-2 / R2-6 / R2-7	Multi-agent, judge, agent-to-agent messages, feedback loop: partly built
	R2-8 / R2-10	Predictive analytics (synthetic data) and AML/KYC/GDPR: partly built

Appendix B: Partial rows in the traceability matrix (2/2)

Area	Row	What is missing (from the traceability Status)
Security	P0-9	Routes added later reuse existing actions
	P0-10	Superuser login; no hash chain
	P0-11	No canary check on the container logs of every service
Operations	P0-12	No tool spans, no exporter
	P0-13	No LLM cost or eval-score metric
	P0-14	No reconciler-unknown rule, no external paging
	P1-5	No production-like load measurement
	P1-6	Graceful shutdown only partly
UI / demo	B13.3	Ops console only partly
	PR-2	Alert → dashboard → runbook demo not recorded
NFR	NFR-SCALE / NFR-AVAIL / NFR-GEO	Design only

The wording here is shortened; `docs/requirements-traceability.md` is the source of truth.

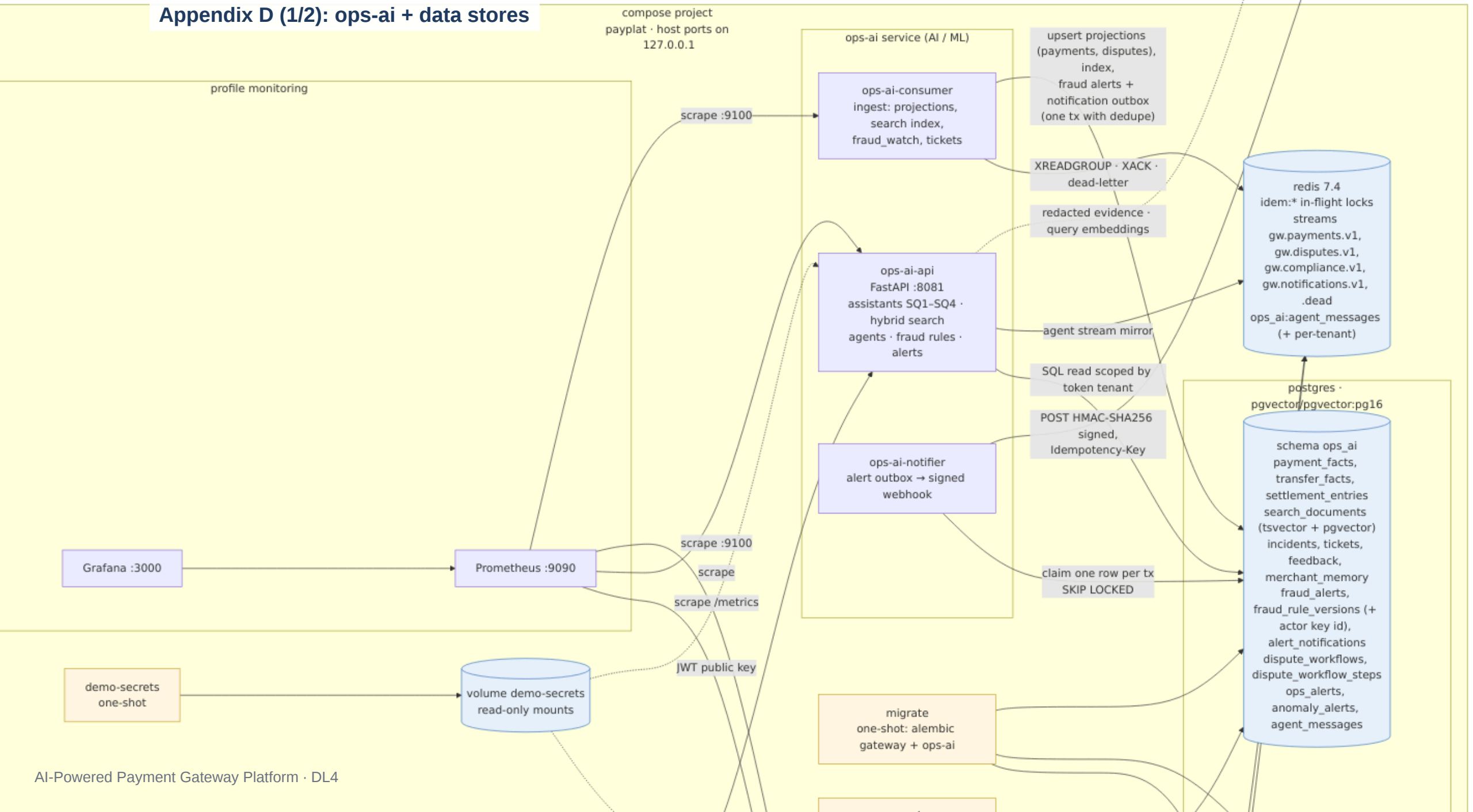
Appendix C: Live demo run-sheet (8:00)

Before (T−15 min, off the clock): `make demo-down && docker compose down -v` → `make demo` **once** (keep the scrollback) → `make web-up`. Tabs: deck, terminal at repo root, `docs/sample-queries.md`.

Start	Screen	Command / what to point at
0:00	Slides 1–2	Scope; architecture
1:15	Terminal + slide 3	`make demo` scrollback: approved / declined / same-key retry (identical body); `all checks passed`
1:50	Slide 4	Tools-first; MockLlm scope
2:30	Terminal + slides 5–8	`docker compose --profile demo run --rm --no-deps -T demo python scripts/sample_queries.py`, then one line per SQ (0:40 each)
		SQ1: 36 / 241, BIN `400000`, `inc_0001` · SQ2: `cross_border_limit_for_kyc_tier`
		SQ3: 2 flagged episodes, `fraud` tickets (optional: `docker compose logs ops-ai-notifier`) · SQ4: `verified: true`, 8 lines
5:15	Slides 9–11	Security, NFR, production design
7:15	Slide 12	Gaps, next steps, summary; Q&A from 7:50

- **If the stack fails** (build, health check or any command): do not debug live; open **`docs/sample-queries.md`** (captured 2026-11-05 from a fresh stack) and keep the timings
- Running `make demo` again adds live charges, so counts can drift from the slides; timestamps always differ
- Q&A only: Appendix A / B; PR-1 (≈ 40 s) on request; web UI `http://127.0.0.1:8082` → Merchant tab (paste the `make web-merchant-key` key **off-screen**)

Appendix D (1/2): ops-ai + data stores



Appendix D (2/2): gateway + clients

